NATQ2-DIAG-001 — why SYSTEM-H missed the floor

Trace-only · 0 systems run · 0 validation runs consumed · validation closed as FAIL, not reopened · reserve 0 bytes · head `1d4a10a` · 2026-09-05 18:54Z

SYSTEM-H did not fail because it could not find the evidence. It found the evidence and ranked it away. Of the 57 gold spans, **44 reached the final candidate pool** and only 32 survived into the top 10. The 12 that were ranked out took **10 whole cases** from hit to miss. Recover only those, changing nothing about retrieval, and `case_hit@10` goes from **23/40 = 0.575** to **33/40 = 0.825** — above the 0.8 PASS floor. This is an *oracle bound on the stored pool, not a prediction*: it establishes where the loss happened, not that any particular fix would recover it.

44 → 32
SPANS IN POOL → IN TOP 10

12
SPANS RANKED OUT

33/40
ANY-SPAN CANDIDATE CEILING

0/57
SPANS FROM PROJECTION

0
RUNS CONSUMED HERE

1 — Where the 57 gold spans were lost

Stage	Spans	Rate
Gold spans in the validation partition	57	—
Gold document present in the union pool	56	56/57
Covering candidate in the final pre-rerank pool	44	44/57
Reached the final top 10	32	32/57
In the pool but ranked out	12	12/44

Only **one** gold span had its document miss the candidate pool entirely. Twelve more had the document but no chunk covering the span — localization. The remaining twelve losses happened after retrieval had already succeeded.

2 — The cross-encoder is the locus

Spans that survived carry a median CE score of **1.2847** at a median CE-score rank of **3.0** in their pool. Spans that were ranked out carry a median of **-3.3642** at rank **20.0**. **11/12** of the ranked-out spans were scored negative. The reranker is not undecided about this evidence — it is confidently against it.

Case	Span	CE-score rank in pool	CE score
A07	1	63	**-2.116**
A30	0	18	**-2.268**
B09	0	20	**-6.37**
B09	1	20	**-6.37**
B22	0	50	**-7.578**
B28	0	17	**-4.518**
C03	0	31	**-2.389**
C09	0	19	0.096
D03	0	17	**-4.226**
D08	0	7	**-3.351**
D27	0	32	**-3.378**
E06	0	30	**-2.796**

One row implicates the blend rather than the cross-encoder: **D08** span 0 sat at CE-score rank 7 and still finished outside the top 10, so the EXP-017/EXP-019A blend demoted it. That is 1 of 12; the other 11 were already far down the CE ordering.

3 — Candidate generation caps the system at 0.825

CANDIDATE_CASE_CEILING, every-span oracle: **31/40**. Candidate span ceiling: **44/57**. Any-span ceiling, which is what bounds `case_hit@10`: **33/40**.

So a *perfect* reranker over the pool as generated reaches 33/40 — one case above the 0.8 floor. Ranking is where the recoverable loss is, but the margin behind it is a single case. Any future work that improves ranking without improving candidate generation is working inside that cap. These are oracle bounds and are not system performance.

4 — The projection stage contributed no gold coverage

> Across the 40 cases the projection stage supplied **800** candidates (20 per case) and covered **0** gold spans. All 44 in-pool gold spans arrived through the SYSTEM-A / local-BM25 fusion. Meanwhile 12 projection chunks did occupy final top-10 slots, at a mean 885.58 ms per case.
>
> It establishes that 800 projection candidates covered zero gold spans, and that 12 of the 400 top-10 slots went to projection chunks that covered no gold span. It does NOT establish that removing the stage would raise case_hit@10 — that requires a counterfactual ranking this analysis is not authorised to produce.

5 — Failure taxonomy

B_LOCALIZATION_FAILURE 7, C_RERANKING_FAILURE 8, D_MULTI_SPAN_PARTIAL_FAILURE 1, F_MIXED 2. No case was left TRACE_INSUFFICIENT.

Case	Population	Spans	Failed	Primary mechanism	Secondary
A07	BOTH_HIT	2	1	D_MULTI_SPAN_PARTIAL_FAILURE	C_RERANKING_FAILURE
A12	BOTH_MISS	3	3	B_LOCALIZATION_FAILURE	—
A30	BOTH_MISS	2	2	F_MIXED	B_LOCALIZATION_FAILURE, C_RERANKING_FAILURE
B09	H_REGRESSIONS	2	2	C_RERANKING_FAILURE	—
B18	BOTH_MISS	1	1	B_LOCALIZATION_FAILURE	—
B19	BOTH_MISS	2	2	B_LOCALIZATION_FAILURE	—
B22	BOTH_MISS	1	1	C_RERANKING_FAILURE	—
B28	H_REGRESSIONS	1	1	C_RERANKING_FAILURE	—
C01	BOTH_MISS	2	2	B_LOCALIZATION_FAILURE	—
C03	BOTH_MISS	2	2	F_MIXED	C_RERANKING_FAILURE, E_DOCUMENT_DISCOVERY_FAILURE
C05	BOTH_MISS	1	1	B_LOCALIZATION_FAILURE	—
C09	BOTH_MISS	1	1	C_RERANKING_FAILURE	—
D03	BOTH_MISS	1	1	C_RERANKING_FAILURE	—
D08	BOTH_MISS	1	1	C_RERANKING_FAILURE	—
D27	H_REGRESSIONS	1	1	C_RERANKING_FAILURE	—
E01	BOTH_MISS	1	1	B_LOCALIZATION_FAILURE	—
E06	BOTH_MISS	1	1	C_RERANKING_FAILURE	—
E27	BOTH_MISS	1	1	B_LOCALIZATION_FAILURE	—

All three **H_REGRESSIONS** (B09, B28, D27) are reranking failures: SYSTEM-H had the covering candidate in its pool and ranked it out, while BM25 found the same evidence and kept it. Even inside BOTH_MISS, five of fourteen are ranking losses rather than retrieval losses.

6 — What these traces cannot say

- system a global pool — UNAVAILABLE — membership not persisted separately from local BM25
- local bm25 additive — UNAVAILABLE — membership not persisted separately from SYSTEM-A
- pre ce rrf rank — UNAVAILABLE — only pool sizes were persisted

Reranker movement could not be computed. Pre-CE RRF rank was never persisted, so no per-span before/after delta exists. The CE-score ranks above are derived by ordering the stored CE scores and are diagnostic only, because the final order is a blend. GLOBAL_CANDIDATE_FAILURE and DOCUMENT_DISCOVERY_FAILURE are likewise not separable here: SYSTEM-A membership was never persisted apart from local BM25, so the one span whose document never entered the pool is reported as E rather than split against a distinction the traces cannot support. Nothing above was recreated by rerunning anything.

7 — Answer

> SYSTEM-H outperformed BM25 because its reranking stage genuinely recovers evidence BM25 cannot surface: 11 rescued cases and a significant paired micro-MRR gain. It nonetheless missed the absolute floor because the same reranking stage discards evidence it already holds. 44 of 57 gold spans were retrieved into the final candidate pool and 12 of them were ranked out, taking 10 cases from hit to miss. Had those 10 held, case_hit@10 would have been 0.825 and above the PASS floor. The failure is therefore located in ranking, and within ranking primarily in the cross-encoder, which scored 11 of those 12 spans negative with a median score of -3.36 against +1.29 for spans that survived. Candidate generation is a secondary constraint that caps the system at 0.825 with only one case of margin.

8 — Artifacts

File	SHA256
NATQ2-DIAG-001-POPULATIONS.json	`fcc8ddb9a3a5d912232b55be2902df09a46eeb9eb394a1900f1ce16d2f85ff4e`
NATQ2-DIAG-001-SPAN-TRACE.json	`1f949aac5cfb23c00f5410bd7704c33d5b66015f253240fe94d77078fdcddd36`
NATQ2-DIAG-001-CEILING-TAXONOMY.json	`3ce530321564163610f4607f3fc35f1a31d813dbc563af1c10cd968438c34026`
NATQ2-DIAG-001-ANALYSIS.json	`30e3acda280c509e8aa4aeae53687b36c12c158d2b285e170c63d85c40031aa1`

Authoritative SYSTEM-H config hash `7599eb3c3bb4798230a722e6a7c96b046dc36cc0332b584ec55339896b2d717a`, unchanged. SYSTEM_H_NATQ2_VALIDATION_CLOSED = true; SYSTEM_H_NATQ2_RETRY_AUTHORIZED = false. Neither remaining validation run was used. No retrieval, reranking, or CE inference was performed; the only external read was the frozen chunk table, to resolve stored chunk ids into offsets they already carried. The task message was truncated mid-sentence in the RERANKER MOVEMENT section; every fully specified section was completed and any section after it was not received. Prior status at head `1d4a10a`.